

CSMYC

Geo-Positioning Buoy Project



In collaboration with Baker Creations, LLC

<https://www.dronebuoyproducts.com/>

Chester Springs Model Yacht Club

<https://sites.google.com/site/csmyclub/>

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John Stoudt

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Geo-Positioning Buoy Project

The Chester Springs Model Yacht Club wanted to acquire buoys that could be positioned without having to go out on the water to place them. The Geo-Positioning Buoy (GPB) is the result of the club's effort to have a buoy that can be set from shore using a hand-held device to position it.

The club established these goals:

1. To have buoys that we could place from shore, the same as our boats.
2. To make our project available to other clubs.

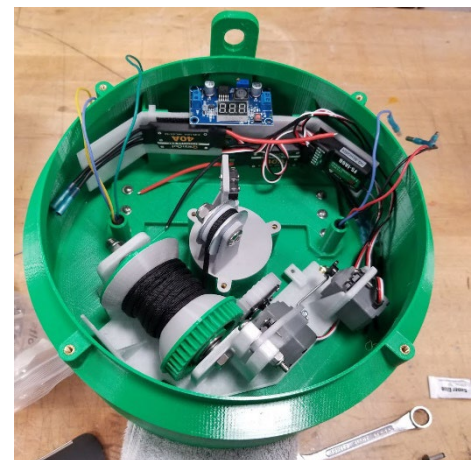
We wanted geo-positioning buoys from the start. We searched the internet for reasonably priced Geo-Positioning Buoys to no avail. One of our club members found an expensive version in Australia. Another member found someone in England making 3D-printed buoys for his club. The UK version carries a weight out to the desired location and drops it. Robin Gray, the developer of these buoys graciously sent us his CADD and 3D printing files. Our team redesigned the mechanism within the buoy to allow it to carry and drop a 3.5-pound weight (anchor). Robin's design carried a 2-pound weight. We knew we need a heavier weight to keep our marks on station at our sailing location.

It took us a few months to redesign this anchor buoy and get it working. Three days after we had successfully bench-tested the anchor buoy, another club member found a more reasonably priced buoy using geo-positioning. We immediately reached out to the owner of the company, David Baker. Our interest was in putting his electronics in our "hull." We arranged to buy one of his kits and began the process of redesigning our buoy once again. A few weeks later we completed the redesign of our Geo-Positioning Buoy (GPB) with the modifications necessary to fit his electronics into our hull.

We are lucky to have members of our club who have access to large 3D printers. These printers are able to print the redesigned buoy with better speed and accuracy. We printed the parts in PETG which is a strong and UV resistant plastic.



GPB Out of the Water



Anchor Buoy Mechanicals

Here is Why Geo-Positioning Buoys!

Geo-positioning buoys are controlled by an App on your phone or tablet using Bluetooth Low Energy (BLE). Pick the buoy you want to control, command its autopilot to move along desired headings, then touch the anchor button. Or you can use stored waypoints and command the buoy to a waypoint. BLE has greater connectivity range than standard Bluetooth with excellent properties over water. Any Bluetooth equipped phone or tablet can control all buoys.



*Buoy Holding Position
in 20 Knot Winds*

Improved Software – Baker Creations has improved the software, calling it Fleet Solution. This new program utilizes a main (hub) buoy and secondary (spoke) buoys.

The spoke buoys can be up to 185 meters (600 feet) away from the hub buoy and still be controlled. The hub buoy can hold up to 40 waypoints. You set up the buoys (individually or as a group) with waypoints to define anchor locations. They can be set up to replicate the courses your club sails. Once the buoys are deployed to a particular course, you have options, if the wind conditions change. You can:

- Shorten or lengthen the course
- Rotate the course clockwise or counterclockwise
- Translate (slide) the course in any direction (including to another venue)
- Send all the buoys to the operator's location (come home command)
- Move an individual buoy, independent of the others.

The buoys can also be controlled with a radio control TX/RX system.

The anchored buoy versus a geo-positioning buoy. These two buoys seem to behave differently. They both react to the wind speed, current, and wind direction. The geo-positioning buoy's movement is more visible because the geo-positioning software reacts more quickly, making the buoy's movement more noticeable. The anchored buoy is tethered to the bottom with a heavy weight and a counter weight that allows the buoy to move downwind as wind speed increases or spin around the bottom weight as the wind direction changes. The movement of the anchored buoy is subtle compared with the geo-positioning buoy. They both move about the same distance. As the wind increases, the anchor rode lengthens. As the wind subsidies, the anchor rode shortens. The movement is the same whether you use an anchored buoy or a geo-positioning buoy.

This movement of the geo-positioning buoy can be controlled by adjusting the RANGE GAIN to -40% and by adding the stability keel.

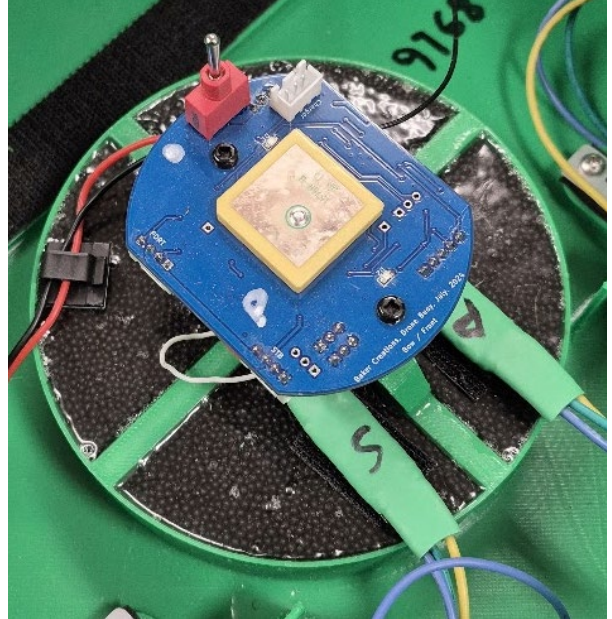
Geo-Positioning Buoy Components

Hull

The GPB hull parts may be ordered from AUTOTIVMFG, Salem, NH. The 3D-printed parts include the hull body, lid, flag mast top button, and the flag base. The purchase price may vary based on quantity and color. The parts will have the heat sink nuts installed when you receive them.

Order the hull parts from AUTOTIV here:

<https://autotiv.myshopify.com/>



Electronics

The electronics package may be purchased from Baker Creations, LLC. You can purchase a short kit if you wish to: <https://www.dronebuoyproducts.com/product-page/byob-build-your-own-buoy>

Inquiries may be sent to: info@dronebuoyproducts.com. All the electrical components needed for the CSMYC GPB design are included in the kit. The short kit includes two thrusters, two electronic speed controls (ESCs), battery and charger, control board, charging cable, and battery extension cable.

Miscellaneous Parts

The miscellaneous parts are noted on the spreadsheet in Appendix 3 (page 19) and are needed to assemble the GPB. In many cases it was easier and more cost effective to buy them in quantity on-line rather than by the piece from a local hardware store. You will notice on the spreadsheet that we have provided you with a cost comparison for one buoy and for six buoys.

NOTE: If an item on the spreadsheet is underlined and in color that means it is an active link. If you click on it with your right mouse button, you will be taken directly to the on-line page for that item. The Excel spreadsheet can be found here – [Parts List](#)
If a link becomes inactive, which can happen, just Google the item description in the search box on the website noted in the key to the spreadsheet.

Optional Stability

Before we tested our GPB, we had designed a “keel” for additional stability. Through the testing process increased stability was written into the firmware; however, you may need more stability due to your sailing location. The keel can improve stability in an area where there is a lot of wave action, current/tides, or high winds. Building a keel is outlined on page 15.

Battery Options

You need a minimum of 8.4 volts to run the thrusters. The control board provides a regulated 5.0 V to the receiver if you wish to control the GPB using a radio/receiver combination. Therefore, there are three options for batteries to power the GPB. Go to Appendix 2 for more complete information.

1. LiPo
2. LiFe
3. NIMH

Digital Files

The 3D printer files are available if you want to print the buoy parts yourself. We used a Creality, Model K1 Max printer, with a print volume of 300mm x 300mm x 300mm. The CADD files were drawn in Free-CAD (<https://www.freecad.org/>) and exported as STL files. Final preparation of the printer files was performed in either OrcaSlicer or Cura; any slicer program will work. The CSMYC will provide the STL files for a donation of \$50.00 to help defray developmental costs. If you would like copies of the STL files, please contact John Stoudt (jstoudt309@gmail.com).

We chose to print the buoy parts from PETG rather than PLA, ABS, or ASA because PETG is a strong material that is UV resistant, while not requiring special fume ventilation while printing. Your printer must be able to manage the PETG filament. We have provided the print parameters for PETG in Appendix 4, on page 20. We do not have print parameters for other materials.

ASSEMBLY PROCEDURES

There are multiple procedures in this manual including the construction of the various components, installation of the electronics, set up, and use. The parts are acquired from various sources including Baker Creations, LLC (electronics), Autotiv (hull components), and from various suppliers (miscellaneous parts) noted on the spreadsheet (Appendix 3 – page 19).

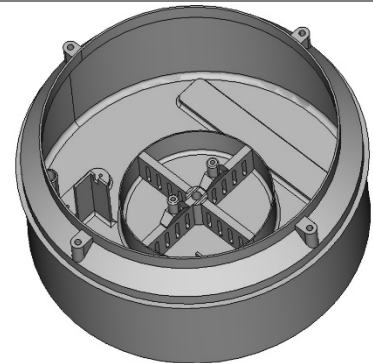
It is suggested that you read the instructions completely and study each section before you begin assembly. It is a rather simple build and should take about two hours to complete one buoy.

You will need one 3D-printed hull, lid, and flag base for each GPB. 3D-printed parts may have some residue left on them that needs to be removed. This residue may be some plastic fiber or “bridging” used in the digital construction of the print file to ensure the quality of the part. Remove all of this print residue.

The purpose of the ballast is to make the hull more stable.

1. Weigh out four, one-pound amounts of #9 lead shot. Small mixing cups work well for this.
2. Carefully pour the shot into each quadrant (pie-shaped areas) within the circular center compartment of the hull.

NOTE: The shot will flow through the vertical slots between the quadrants. These slots allow the shot and epoxy to flow between the quadrants and lock the mass into place once the epoxy has set.



CADD Drawing of the Hull

3. Take a ½” dowel and, using the end, lightly tap on the shot in each quadrant to compact it and move it around evenly between the quadrants. This will help the shot to level off and equalize the weight between the four areas.
4. Using the dowel, lightly tap the area of the hull around the outside of the compartmented area. This will level out the lead shot in the compartments.
5. Put two pumps of the epoxy and hardener into the mixing cup and stir for one minute. Use West System epoxy!

NOTE: It is recommended that you buy the pumps for the epoxy containers (West System). They measure out the exact amount of each part of epoxy with each full plunge of the pump handle.

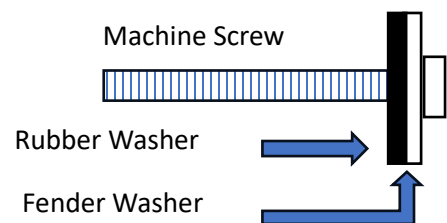
6. Slowly pour some of the mixture into each compartment being careful to keep it from overflowing the edges of the compartments.
7. This process will take a total of eight pumps of the mixture to completely cover the shot in all the compartments. More or less epoxy may be needed. Watch the epoxy “settle” into the shot and eventually level out in each compartment. Take your time adding small amounts of the mixture to each compartment. According to our calculations the mass of the shot and epoxy will just fill the four compartments, and the epoxy will cover the shot completely.



Ballast Filled Compartments

The purpose of the keel plug is to plug the hole in the bottom center of the hull to keep the hull watertight.

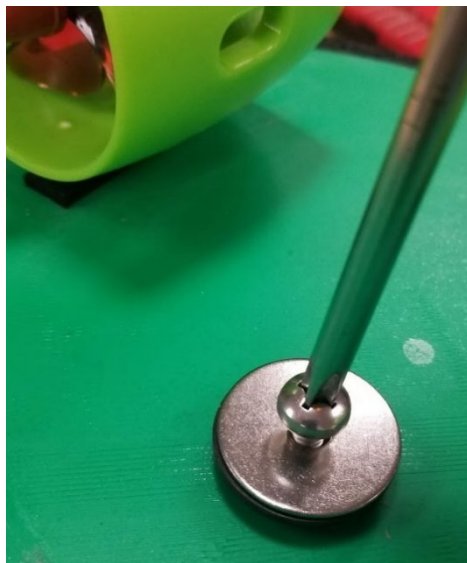
1. Place the $\frac{1}{4}$ " x $1\frac{1}{4}$ " stainless steel fender washer on the $\frac{1}{4}$ " – 20 x $1\frac{1}{2}$ " stainless steel round head machine screw. Then place the $\frac{1}{4}$ " x $1\frac{1}{4}$ " rubber washer on the machine screw.
2. Install the assembly from the bottom of the hull, threading it into the $\frac{1}{4}$ " – 20 brass heat insert nut in the hull. When inserted, you can see the tip of the machine screw in the bottom center of the inside of the hull.
3. You may have to drill out the $\frac{1}{4}$ " hole from the bottom to remove any plastic residue. Carefully drill into the hole with a $\frac{1}{4}$ " drill bit, until you feel the drill hit the brass nut insert.



Keel Plug

NOTE: Sometimes the plastic gets pushed into puddles in the holes where the heat sink nuts are installed. Some of this plastic may accumulate in the threads. Carefully drill out the access plastic using appropriately sized drill – the pilot hole size of the thread for $\frac{1}{4}$ "

– 20 TPI is a #7 drill. On occasion you may have to run the correct tap through the threads to clean them up (1/4" – 20 TPI).



Keel Plug Being Installed

The thrusters provide the propulsion to the GPB.

NOTE: The boxes the thrusters come in are marked CCW (counter clockwise) and CW (clockwise). Mark the thrusters accordingly. They are mounted as follows:

- a. CW on the starboard side.
- b. CCW on the port side.



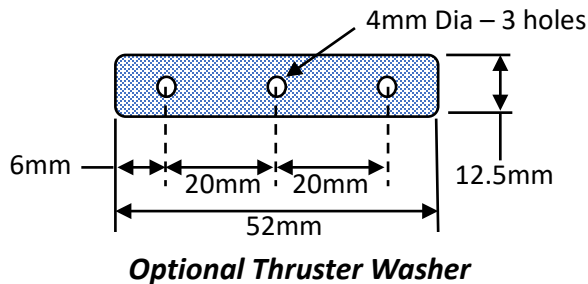
Thrusters Before Installation

1. If necessary, drill out the six thruster mounting holes on the hull with a 4mm (3/16") drill.
2. Cut four pieces of the 1/8" x 1/2" foam weather strip 52mm long.
3. Adhere the four pieces of foam, centered on the three holes, on the bottom and inside the hull for each thruster.
4. Press them down firmly to make sure the adhesive adheres to the hull surface.
5. Take an ice pick or narrow awl and poke a hole in the foam at each hole location, on the inside of the hull, to give the screws a place to start. This will ensure the weatherstripping does its job sealing off the areas around the holes. Do not drill through the foam, as you want a tight seal.

6. Place a 4" piece of a suitable size piece of shrink tubing on the three wires (it will just fit over the wires) coming out of each thruster. Slide it down tight against the thruster body. Using a heat gun "shrink" the tubing onto the wires.
7. Slide "O" rings over the shrink tubing in the following order:
 - 11mm OD, 4mm ID, 3.5mm width
 - 10mm OD 3mm ID 3.5mm width
8. Take one thruster and feed wires through the wire access tube in the hull.
9. Place a #6 stainless steel washer on each screw. Or you may custom make a three-hole washer, like the diagram, out of 1/16" x 1/2" aluminum. We found this a more effective way to



"O" Ring Awaiting Sealant

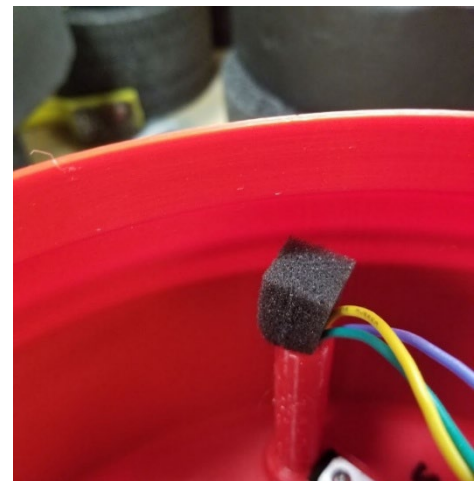


ensure the seal around the gasket.

10. Insert the six #6 x 1" stainless steel pan head sheet metal screws, through the washers, into the weatherstripping in the six-hole locations. Slowly start to thread them into the weatherstripping by hand, through the hull and into the correct hole in the thruster.

NOTE: The thrusters are facing the correct direction if the wires go directly into the hole in the hull above where the wires come out of the thruster.

11. Finish tightening each screw with a screwdriver. You will see the gaskets flatten and spread out under the pressure.
12. Push the two "O" rings into the hole in the bottom of the hull and recess them 1/4". Seal the area with a small amount of clear silicon sealant. If you forgot to put the gaskets on before mounting the thruster, you will need to pull the wires out of the access tube, install the gaskets, then re-feed the wired though the access tube.
13. Repeat this process for the other thruster.
14. Cut a piece of foam rubber 1/2" x 1/2" x 2" (open cell works best for this application) and carefully stuff it into the hole.



Foam Being Inserted into the Tube

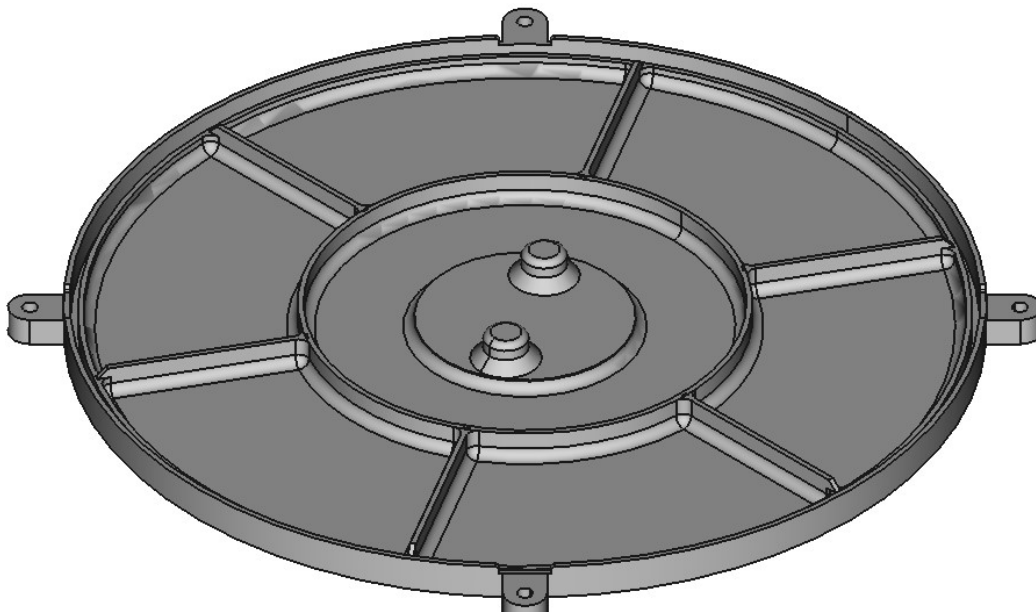
15. Place two “O” rings over the wires and work it into the top of the tube. Do it in this order:
 10mm OD, 3mm ID, 3.5mm width
 11mm OD, 4mm ID, 3.5mm width
16. Repeat this for the other thruster.

The purpose of the lid gasket is to keep the Geo-Positioning Buoy watertight.

1. The lid has a groove on its bottom outside edge. The purpose of this groove is to receive a gasket called the lid seal.
2. Starting at some point along the groove take the 5/64” round rubber weather seal and push it into the groove.
3. Wait until you have completed the installation to cut the seal to length.
4. You can push the seal into place using a #2 straight screwdriver.
5. Do not slide the screwdriver along to install the seal gasket. This will stretch the gasket material and make the seal too thin in places.
6. Just keep pushing the gasket into the groove with the blade of the screwdriver.
7. When you have reached the end, trim it off with a small pair of scissors and push the ends into the groove. No adhesive is necessary.



Installing the Gasket



CADD Drawing of the Lid (underside)

The purpose of the rub rail is to protect boats and Geo-Positioning Buoy from damage and to provide more stability.

1. The rub rails are cut from a sheet of ½” x 12” x 54” adhesive-backed closed cell foam rubber. Closed cell will not absorb water like a sponge thus providing buoyancy.
2. Carefully cut this sheet lengthwise into 3” wide strips. You will need a long metal straight edge and a sharp utility knife. Cutting these pieces must be done with care using a sharp blade in your utility knife. It is strongly recommended that you use a new blade for each strip. This will minimize the tear out along the cut edge. We found the Kobalt Serrated Utility Blades (P# 2491092) to do a very nice job!
3. Cut one of these strips 40” long.
4. Install this piece along the bottom edge of the hull, flush with the bottom. You will need to manage the tape backing as you peel it away.
5. The piece of foam will be too long and needs to be trimmed.
6. When you know where the pieces join, carefully cut the end to fit against the start end of the rubber. Cut it a little long and then trim it to its final length.
7. Roll the GPB on its edge compressing the foam to ensure a good adhesion to the hull and applying pressure to the adhesive.
8. Pull the joint apart with your thumb and forefinger. Work GOOP (adhesive) into the joint with a toothpick. When you are satisfied there is enough adhesive in the joint and not too much, allow the joint to go back together.
9. Work the joint with your fingers to ensure the glue seals the entire joint. Wipe off access adhesive.



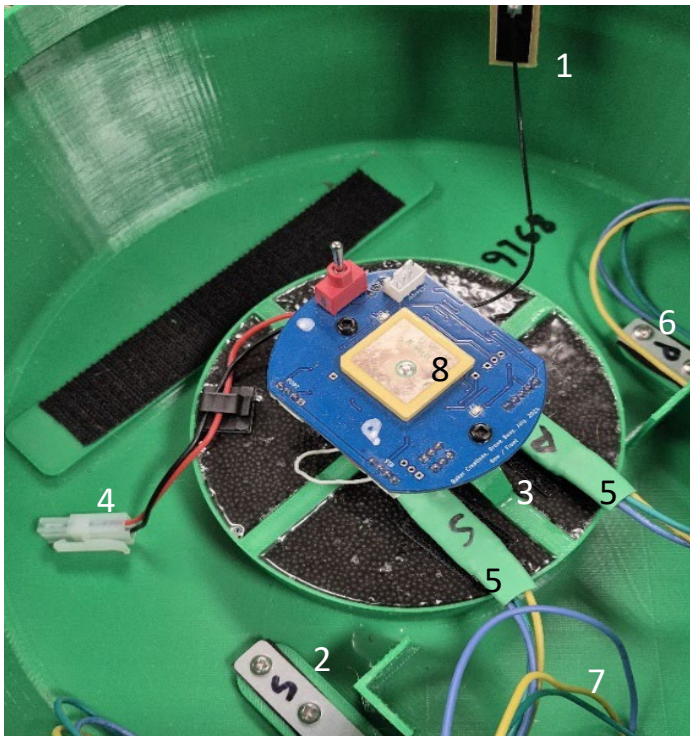
Rub Rail Installed

The purpose of the electronics is make the Geo-Positioning Buoy function without the need for mechanical components other than the thrusters.

NOTE: The electronics package that we used from Baker Creations included the “fleet solution”. This new solution (program) uses a main (hub) buoy and five secondary (spoke) buoys. The number of buoys is controlled by your course setup. You will buy one electronics kit for the “hub” buoy with two Bluetooth antennas and as many other

electronics kits for the “spoke” buoys with one Bluetooth antenna as you need. These components are programmed to work together.

There are four electronic components in the GPB: the computer control board with Bluetooth antenna/s, ESCs (electronic speed control), the thrusters, and battery. An R/C receiver and radio transmitter are an option for control; however, you cannot use the fleet solution with an R/C receiver and transmitter.



Internal Hull Details

You will notice the Internal Hull Details:

1. Bluetooth antenna location
2. Port (P) and Starboard (S) notes
3. Bow indication on the Computer Control Board (CCB)
4. Battery platform and plug wire (special end)
5. Green ESC covers
6. Thruster mounting details
7. Wire management
8. GPS Patch Antenna

WARNING: Avoid touching the top of the GPS patch antenna (#8) while handling the CCB. The GPS receivers

have sensitive RF pre-amplifiers connected to the GPS antenna and can be damaged from static electric discharge.

1. Make sure the wires from the ESCs and the thrusters are all stripped to 3/8”.
2. The wires are color coded to ensure correct operation, but the colors are not always connected as you would think. Start by connecting the same colors with wire nuts.
3. Twist the ends of the wires to ensure there are no loose ends and thread on a small wire nut. This is temporary until you are sure the thrusters and ESCs are connected and working correctly.
4. Plug in the ESC connectors into the bottom of the CCB.
5. Install a power extension cord for ease of plugging in the battery. Make sure you have an extension that matches the connectors on the CCB and your battery.

6. Test for correct operation - Throw the on-off switch to turn on the circuit board. The switch has three positions, only one position (to the right if you are facing the bow) provides power to the circuit board. You should see the blue GPS LEDs light up and hear the motors spin up and back down. The starboard thruster spins first, then the port thruster, then the starboard again, and finally the port. See if the thrusters are connected correctly via this sequence. If they are not, switch the ESC (4-pin) connections at the circuit board and test again.
7. Make sure the motors are thrusting forward. The motors should push air out on the side where the propeller blades are most visible. (Opposite side from the wiring.) If a motor is running backwards, switch two of the three wire connections between the motor and the ESC.
8. Once proper operation is established, turn the circuit board off with the switch.
9. Insert one color set of wires at a time from the ESC and the thruster into a solder seal wire connector. Heat with a heat gun until the shrink wrap insulation shrinks around the wires and the solder melts and joins the wires together. You can watch the solder melt and flow into the wire ends. You now have a solid, watertight connection.
10. Repeat this for the other five sets of wires, one set at a time. Retest the start up.
11. Place two 1/2" x 3/4" Velcro strips on the ballast, on either side of the posts the CCB mounts on, towards the bow. (hook side).
12. Place one 1/2" x 3/4" piece of Velcro on one side of the ESCs. (loop side)
13. Install the ESC boards onto these strips on the ballast.

SUGGESTION: Mark all of the parts and the various mounting locations of the parts with port (P) and starboard (S) indicators. Figure out how you want to arrange your wires in the hull. You must keep everything below the control board for efficient access to the satellite signals. See Internal Hull Details on previous page.

You may use wire management clips to help with the wire management. We did.

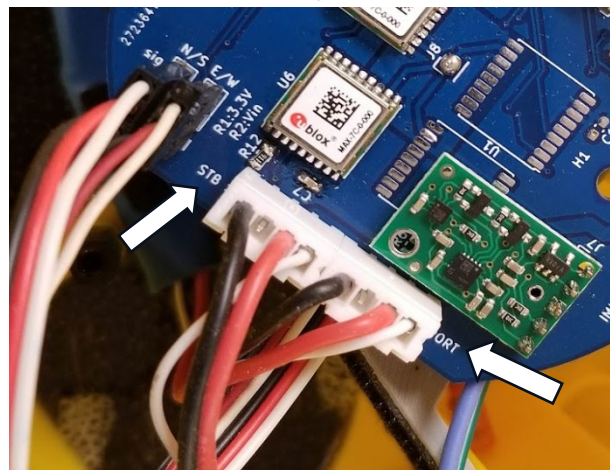


Control Board (arrow points to the bow)

14. Plug the ESC wires into the correct location on the CCB.
15. There are two antenna risers in each buoy hull on either side of the battery riser. Adhere the Bluetooth antenna, or the two antennas in the case of the hub buoy, to the top of the risers using the double-sided tape provided.

NOTE: This is a Bluetooth antenna – it needs to be as high as possible in the hull without interfering with closing the lid, remain vertical, and be at least 2” away from metal. Thus, its location.

16. Using two M3 x 5mm nylon machine screws, secure the control board to the two posts in the center of the hull above the ballast. Note the bow indicator on the control board and place that forward (towards the thrusters).
17. Place a 1” x 4” piece of Velcro on the battery platform (hook side).
18. Place a 1” x 4” piece of Velcro on the battery (loop side).



Note the STB and PORT Indications

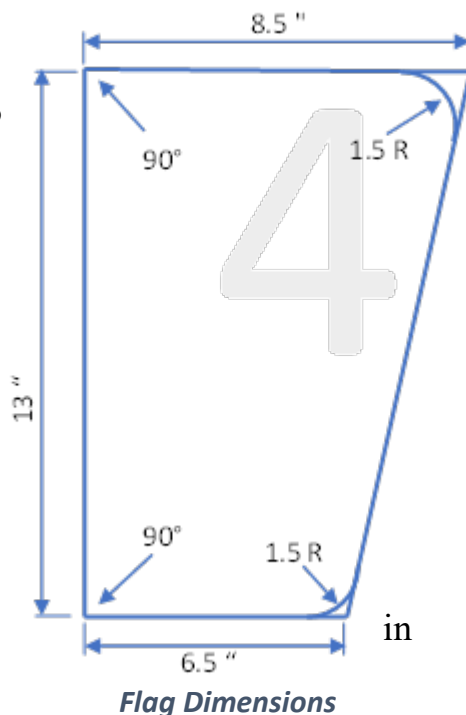
NOTE: The thrusters are mounted towards the bow of the GPB. The ESC plugs are labeled (white arrows) on the bottom of the control board as starboard (STB) and port (PORT). The top of the control board has one end labeled bow (white arrow).

You will also notice where the R/C receiver wires plug into the board (if used).

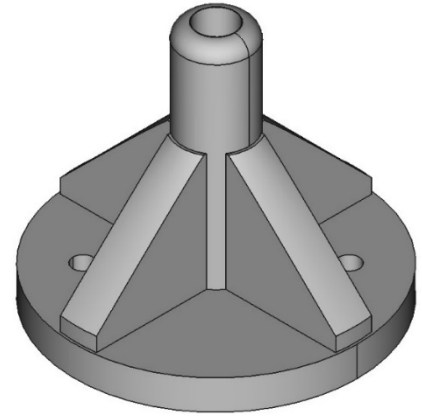
The purpose of the flag assembly is to make the buoy more visible in the water and to give the buoy a specific designation for course position recognition.

Cut a piece of the ¼” diameter orange fiberglass driveway marker 16” long. Sand a bevel on each end. Be careful not to get a fiberglass splinter.

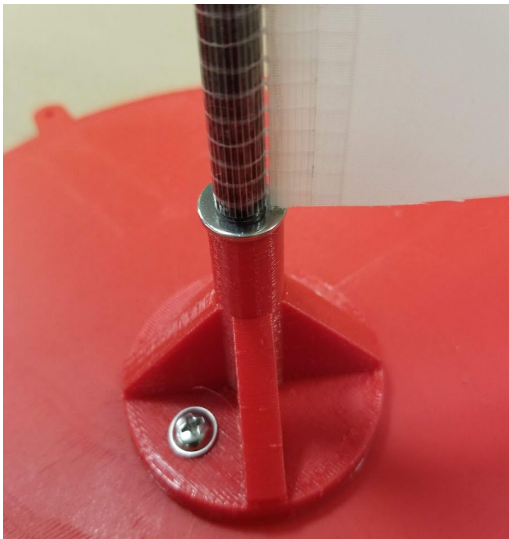
1. Cut a piece of aluminum tubing 13” long and clean up the ends.
2. Cut out a cardboard template in the shape of the flag. See diagram.
3. Trace and cut out as many flags as you need from the ripstop nylon.
4. Add the flag designation to the flag using an indelible marker.
5. Cut a piece of the clear duct tape 13” long.
6. Place the tape upside down on a flat surface. Place the 13” edge of the flag, flush at each end of the tape, ½” from that edge.



7. Take the tube and line it up parallel with the edge of the tape and the flag but not on the tape at this point.
8. Carefully roll the tube towards the tape, allowing the tape to wrap around the tube as you continue to roll the tube.
9. As the tape completely wraps around the tube continue to roll the flag until the tape adheres to the other side of the flag.
10. Now roll in the other direction so you wrap back over the flag. This will allow the flag to stand out “perpendicular” to the tube. Trim off any excess tape.
11. The hole through the center of the flag base is stepped. The top of the hole is a $\frac{17}{64}$ ” diameter, and the lower portion $\frac{1}{4}$ ” diameter (last $\frac{1}{4}$ ”). Run a $\frac{1}{4}$ ” drill bit through the lower portion from the bottom of the base to “open” up the hole.
12. Place the shaft into the flag base from the top. You may have to tap it home with a hammer until it is flush with the bottom of the base.
13. Slide a $\frac{1}{4}$ ” stainless steel washer down over the shaft.
14. Slide the flag/tube assembly over the shaft with the wide end of the flag at the top.
15. Take the “button” and drill out the center hole to $\frac{1}{4}$ ”. Tap the small hole on the side with a 10 – 24 tap.
16. Place the “button” over the top of the flag mast, flush with the top, the bevel down, and insert and tighten a $\frac{1}{8}$ ” x 10- 24 set screw (hardware store).



CADD Drawing of the Flag Base.



Flag Assembly at Its Base



Button Installed on the Mast

A weighted keel adds additional stability (optional) to a Geo-Positioning Buoy.

If you need added stability, you can make a weighted keel rather easily. Its simple design makes it easy to attach a $\frac{3}{4}$ -pound weight that can be moved up and down the keel shaft.

NOTE: The hull is provided with a $\frac{1}{4}$ " hole in the bottom that gets plugged with a $\frac{1}{4}$ " rubber washer, a stainless fender washer, and a $\frac{1}{4}$ " – 20 x 1 $\frac{1}{2}$ " round head stainless steel machine screw. There is a heat sink nut at the top of the hole. The bolt assembly may be removed to install this keel.



Muffin Tin

Assembly

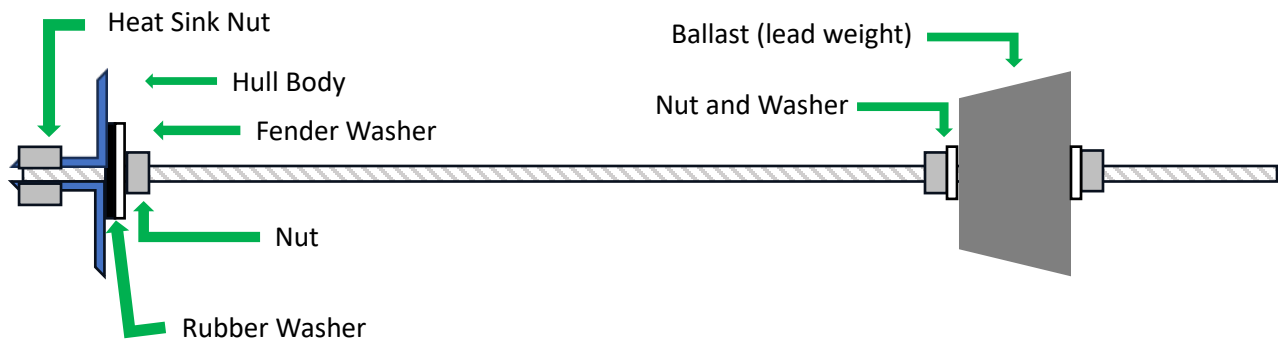
1. Draw a 1 $\frac{3}{8}$ " circle on the piece of cardboard with a compass.
2. Push a hole through the center of the circle with an awl.
3. Cut the disk out.
4. Place the muffin tin on wooden workbench.
5. Drop the cardboard disk into bottom of one of the muffin molds and using a steel shank awl punch a hole into the metal bottom of the mold.
6. Do this in the bottom of each muffin mold.
7. Place the muffin tin on a metal or other hard surface and hammer each hole to flatten the burr around each hole with a small ballpeen hammer.
8. Drill out each of these holes out using a $\frac{3}{32}$ " drill bit.
9. Cut at least 12, 2" pieces of the $\frac{1}{4}$ " dowel.
10. Mark the center of one end of each piece of dowel and drill a $\frac{1}{16}$ " hole, $\frac{1}{2}$ " deep.
11. Using the #2 x $\frac{3}{8}$ " pan sheet metal screws fasten one dowel in the bottom of each muffin mold.



Center Pins Installed (holes shown)

Now for the neat part – As you pour the molten lead into each mold cavity the dowel burns slightly (chars). When the lead cools and you unscrew the screw, the dowel and lead weight release together. Take a pair of pliers and twist the dowel slightly, it turns right out of the casting. This leaves a hole slightly smaller than $\frac{1}{4}$ " in the casting, making it much easier to drill out the center hole to the desired $\frac{1}{4}$ " diameter. You can use the dowels over again one or two times as needed.

12. Set the mold on a heat-resistant surface and pour enough lead into the muffin mold to just fill the mold to the top. Each of these will give you (approximately) a $\frac{3}{4}$ -pound weight. It is suggested that you make two per buoy.
13. You can now assemble one of your “keels” which are made up of:
- 12” piece of $\frac{1}{4}$ ” – 20 stainless steel all-thread
 - $\frac{1}{16}$ ” x $\frac{1}{4}$ ” ID x 1 $\frac{1}{4}$ ” OD rubber washer
 - $\frac{1}{4}$ ” stainless steel fender washer
 - $\frac{1}{4}$ - 20 stainless steel nuts (2 each)
 - $\frac{1}{4}$ ” stainless steel washer (2 each)
 - cast lead weight (you may use more than one)
14. Remove the $\frac{1}{4}$ ” 20 x 1 $\frac{1}{2}$ ” stainless machine screw from the bottom of the hull.
15. Make sure the rubber washer, fender washer and upper nut are down about 1” from the top of the all thread.
16. Push the all-thread into the $\frac{1}{4}$ ” hole in the bottom of the hull until it hits the heat sink nut. Thread it into the heat sink nut until it is flush with the inside top edge of that nut.
17. Slide the rubber washer and fender washer up against the hull and tighten the nut by hand, snugging it with an open-end wrench.



Appendices

Appendix 1 - Key for the [Spreadsheet](#) (Appendix 3)

A – Amazon

BS – box store item

HS – hardware store item

JD – Jamestown Distributors

RM – RotoMetals

Underline – the item has a link to the product on-line

WAMB – this quantity will accommodate multiple BuoyBots

Appendix 2 – Battery Options

A voltage of 8.4V is used to run the thrusters through the GPS controller board. The control board regulates the power out to the receiver if you wish to control the Geo-Positioning Buoy using a radio/receiver combination. Therefore, there are three options for batteries to power the Geo-Positioning Buoy.

LiPo

Board development was based on the use of LiPo cell technology, using a 2S battery to power the buoy system. Because LiPo cells are charged up to 8.4V with a nominal rating of 7.2V, and these batteries are used in electronics for other buoy applications, using 2S LiPo batteries are sufficient for the Geo-Positioning Buoy.

It is suggested that a battery of at least 5000mAh capacity should be used to guarantee at least four hours of continuous use.

Battery Options:

- https://hobbyking.com/en_us/zippy-compact-5000mah-2s1p-40c.html
- <https://www.amazon.com/OVONIC-5200mAh-Connector-Airplane-Helicopter/dp/B07JJ4Q65Z/>

LiFe

While LiFe batteries are lower voltage than LiPo, it is proposed to use these batteries as a safer option than LiPo batteries. The nominal voltage is lower than LiPo, so it would be prudent to come up with a solution that is a higher capacity than 5000mAh.

It is suggested to connect 2 or 3 LiFe batteries in series to obtain a higher total capacity and be able to power the buoy in higher winds or current.

Battery Options:

- https://hobbyking.com/en_us/zippy-compact-4200mah-3s1p-30c-lifepo4-pack.html

- https://hobbyking.com/en_us/turnigy-nano-tech-2000mah-2s1p-20-40c-lifepo4-transmitter-pack.html
- https://hobbyking.com/en_us/turnigy-nano-tech-2100mah-2s1p-20-40c-lifepo4-receiver-pack.html

NIMH

NiMH batteries can be used at the developed board voltage with the nominal rating at 8.4V in a 7-pack of batteries. This will provide a longer run time at the 8.4V rating, and these batteries are safer to charge and use around water. A nominal 8.4V NiMH pack can have a lower capacity rating, because the voltage rating is higher than a LiPo pack. An adapter cable will be needed to used NiMH batteries.

- [We used this NiMH battery.](#)
- [We used this charger for the battery.](#)

Other Options

Custom – 7-cell, AA NiMH packs without connectors can be bought and packaged together to make higher capacity batteries.

A lower-cost NiMH charger can be used with these packs.

- <https://power.tenergy.com/tenergy-8-4v-3800mah-flat-nimh-battery-pack-for-airsoft-rc-car-hobbico-electristar-plane/> (431g)

Examples of discharge curves: http://access.time.angelfire.com/battery_test_data_01.html
Scroll down to “Sheet 2: Analysis “for comparisons.

Alternatively, higher voltage batteries, 3S LiPo or LiFe, or 12-volt NiMH packs, could be used to power the buoy system if a Step-Down Voltage Module is used. An example of a module is here: <https://www.amazon.com/Songhe-LM2596S-LED-Voltmeter-Adjustable/dp/B085WC5G8N/>

These modules are often called an Adjustable Buck Converter, which come in many varieties.

Appendix 3 – Comprehensive Parts List

	Quantity	Item	Details	Key	Each	Cost	Per 1	Per 6 GeoBots		
Miscellaneous Parts for Hull Assembly										
H1	4	Lid knob	M4 x 16 Knurled Knobs	WAMB, A	\$0.28	\$1.12	\$5.49	20 for \$5.49 (2 pkg)	\$10.98	
H2	2	Flag base screw	M4 x 12mm Truss Head Machine Screws	WAMB, A	\$0.24	\$0.48	\$6.99	30 for \$6.99	\$6.99	
H3	2	Control board screw	M3 x 5mm Nylon Phillips Pan Head Machine Screws	WAMB, A	\$0.10	\$0.20	\$9.99	100 for \$9.99	\$9.99	
H4	2	Standoff Screw	M3 8+6mm Standoff spacer screw pillar	WAMB, A	\$0.08	\$0.16	\$7.99	100 for \$7.99	\$7.99	
H5	1	Hull plug	1/4 in-20 X 1-1/2 in Round Head SS Machine Screw	HS	\$1.25	\$1.25	\$1.25	6 each	\$8.50	
H6	1	Hull plug gasket	1/16" x 1/4" ID x 1 1/4" OD rubber washer	HS	\$0.80	\$0.80	\$0.80	6 each	\$4.80	
H7	1	Hull plug washer	1/4" stainless steel washer	WAMB, A	\$0.07	\$0.07	\$6.99	40 for \$6.99	\$6.99	
H8	6	Thruster screw	#6 x 1" Stainless Steel Pan Head Sheet Metal Screw	WAMB, A	\$0.09	\$0.54	\$9.35	100 for \$9.35	\$9.35	
H9	2	Receiver extension	3-Pin Servo Extension Lead (optional for receiver)	WAMB, A	\$0.71	\$1.42		12 for \$8.49		\$8.49
H10	1	Power extension	Power extension cord (optional for LiPo baattery)	WAMB, A	\$1.60	\$1.60	\$6.99	5 for \$6.99		\$13.98
H11	4	Thruster gasket	1/8 x 1/2 x 3" Foam Seal Tape	WAMB, A	\$0.06	\$0.24	\$11.99	50' for 11.99	\$11.99	
H12	1	Lid gasket	5/64" Foam Rubber Weather Seal Strip	WAMB, A	\$1.00	\$1.00	\$6.19	26.25' for \$6.19	\$61.90	
H13	1	Rub rail	36" piece 3 Inch x 1/2 Inch Weather Stripping	WAMB, A	\$7.65	\$7.65	\$22.95	11' for 22.95 (2 pkg)	\$45.90	
H14	6	Solder wire connectors	Solder Seal Wire Connectors	WAMB, A	\$0.16	\$0.96	\$7.99	50 for \$7.99	\$7.99	
H15	4	O rings to seal tubes	Rubber "O" Rings (large)	WAMB, A	\$0.12	\$0.48	\$6.99	50 for \$6.69	\$6.99	
H16	2	O rings to seal tubes	Rubber "O" Rings (small)	WAMB, A	\$0.14	\$0.28	\$6.49	50 for \$6.49	\$6.49	
H17	2	Wire tube inserts	1/2 x 1/2 x 2 foam rubber	scrap						
H18	3	1 x 4 Velcro strips (hook)	Velcro Strips	WAMB, A	\$2.00	\$6.00	\$8.85	8 for \$8.85	\$8.85	
H19	6	Wire Clips	Adhesive Cable Clips (optional)	WAMB, A	\$0.08	\$0.48	\$7.95	100 for \$7.95		\$7.95
H20	6	Stainless Washer	Thruster Washer (optional)	WAMB, A	\$0.09	\$0.54	\$8.99	100 for \$8.99	\$8.99	
H21	1	Sealant	Wire Tube Sealant	WAMB, A	\$1.50	\$1.50	\$7.78	1 tube for \$7.78	\$7.78	
Ballast Materials										
B1	6	West System Epoxy - R	105 Resin - quart	JD	\$1.55	\$9.30	\$49.47	quart for \$49.47	\$49.47	
B2	6	West System Epoxy - H	206 Hardener, must be slow cure	JD	\$0.95	\$5.70	\$30.88	pint for \$30.88	\$30.88	
B3	1	West System Pumps	West System Pumps	JD	\$26.79	\$26.79	\$26.79	set for \$26.79	\$26.79	
B4	1	Stirring stick	Stirring sticks	WAMB, A	\$0.02	\$0.02	\$4.99	200 for \$4.99	\$4.99	
B5	1	Mixing cup	Graduated mixing cups	WAMB, A	\$0.21	\$0.21	\$10.37	50 for \$10.37	\$10.98	
B6	4	Lead shot	#9 Lead Shot (pounds per buoy)	WAMB, RM	\$2.64	\$10.56	\$68.98	25 pounds for 68.98	\$68.98	
Flag Materials										
F1	1	Shaft Lock	Faucet Washer	WAMB, A	\$0.03	\$0.03	\$6.19	25 for \$6.19	\$6.19	
F2	1	Flag mast	15 1/2" piece of 1/4" Orange Fiberglass Marker	BS, HS	\$0.50	\$0.50	\$2.00	2 for \$2.00	\$4.00	
F3	1	Flag swivel	5/16 ID x 13" aluminum tube or arrow shaft	WAMB, A	\$2.10	\$2.10	\$28.75	4 for \$28.75	\$28.75	
F4	1	Flag adhesive	2" x 13" clear duct tape	WAMB, A	\$0.30	\$0.30	\$11.88	1 for \$11.88	\$11.88	
F5	1	Nuts	See K4 below		\$0.09	\$0.09		6 each	\$0.54	
F6	1	Flag body	Ripstop Nylon Sail Fabric – 1 yard, white	WAMB, A	\$1.50	\$1.50	\$8.05	1 for \$8.05	\$8.05	
F7	2	Base bolt	M4 x 12mm Truss Head Stainless Machine Screw	WAMB, A	\$0.33	\$0.66	\$6.99	30 for \$6.99	\$6.99	
					Basic GeoBot Cost		\$84.53	\$407.35	Total for 6 Bots	\$490.96
Keel Materials (optional)			Keel Materials (optional)							
K1	1	Keel weight mold	mini-muffin tin	WAMB, A	\$12.93	\$12.93	\$12.93	1 each	\$12.93	
K2	1	Keel shaft	1/4" – 20 x 12" stainless steel all-thread	WAMB, A	\$6.50	\$6.50	\$12.99	2 for 12.99	\$25.98	
K3	2	Retainer washer	1/4" stainless steel washer	WAMB, A	\$0.07	\$0.13		see H6 above		
K4	3	Nut	1/4" – 20 stainless steel nuts	WAMB, A	\$0.09	\$0.27	\$8.68	100 for \$8.68	\$1.62	
K5	1	Fender washer	1/4" stainless steel fender washer	HS	\$0.80	\$0.80	\$0.80	6 each	\$4.80	
K6	1	Washer gasket	1/16" x 1/4" ID x 1 1/4" OD rubber washer	HS	\$0.80	\$0.80	\$0.80	6 each	\$4.80	
K7	12	Small screws	#2 x 3/4 pan head sheet metal screw	WAMB, A	\$0.10	\$1.20	\$10.02	100 for \$10.02	\$10.02	
K8	1	Disk template	3 inch square of (cardboard) tablet backing	Scrap						
K9	5 to 6	Bulk lead (weights)	Pounds of lead for casting weights	Scrap						

Appendix 4 – 3D Print Notes and Parameters

Our printing parameters were:

Material: PETG

Nozzle: 0.4mm

Layer height: 0.2mm

Infill: 100%

Wall Ordering: Outside to Inside

Requires a heated bed of 300mm x 300mm x 130mm minimum.

A closed chamber printer is also highly recommended.

We used a Creality K1 Max. Our PETG was Hatchbox brand.

Appendix 5 - Buoy Fleet Colors and Designations

The CSMYC has been using drop marks (buoys) for many years. Our start/finish marks are green, the windward marks yellow, and the leeward marks red. We give each buoy a designation on its flag. The start/finish buoy flags are marked with an “S”, the yellow buoys with a “1” and a “2”, and the red buoys with a “3” and a “4”. Once our boats have crossed the start line, we round the buoys in numerical order. Using the fleet solution from Baker Creations, we further mark the GPBs with the following designations on their lids and hulls:

- Starboard Start Buoy – SS (this is the hub buoy)
- Port Start Buoy – PS
- Windward 1 – W1
- Windward 2 – W2
- Leeward 3 – L3
- Leeward 4 – L4

The buoys always point into the wind, so we place a white stripe on the lid that goes from the flag base to the bow of the buoy.

Appendix 6 – Cost Estimates

The costs vary based on the number of buoys that are built at one time.

These are priced per buoy in the quantity noted and the estimated costs at the time of the posting of this document.

Components	Supplier	1 Buoy	2 Buoys	6 Buoys
Hull Components	Autotiv	\$290	\$210	\$160
Electronics	Baker Creations	\$219	\$219	\$219
Misc Parts	Various	\$407	\$204	\$82
TOTAL		\$916	\$633	\$461

Shipping and handling, where applicable, are extra.

Hull Components - Hull, lid, and flag base.

Electronics Components - All the electrical components needed for a GPB, including two thrusters, two ESCs, battery and charger, control board, charging cable and battery extension cable.

Miscellaneous Parts - All other parts as itemized in the parts list (page 19).

NOTE: We priced this out knowing that we would build six GPBs. The cost for the parts for one buoy using this method is excessive. You may be able to source a smaller number of parts locally or by other means. The descriptions provided should enable you to find other sources. It is more economical to build an entire fleet than just one buoy.